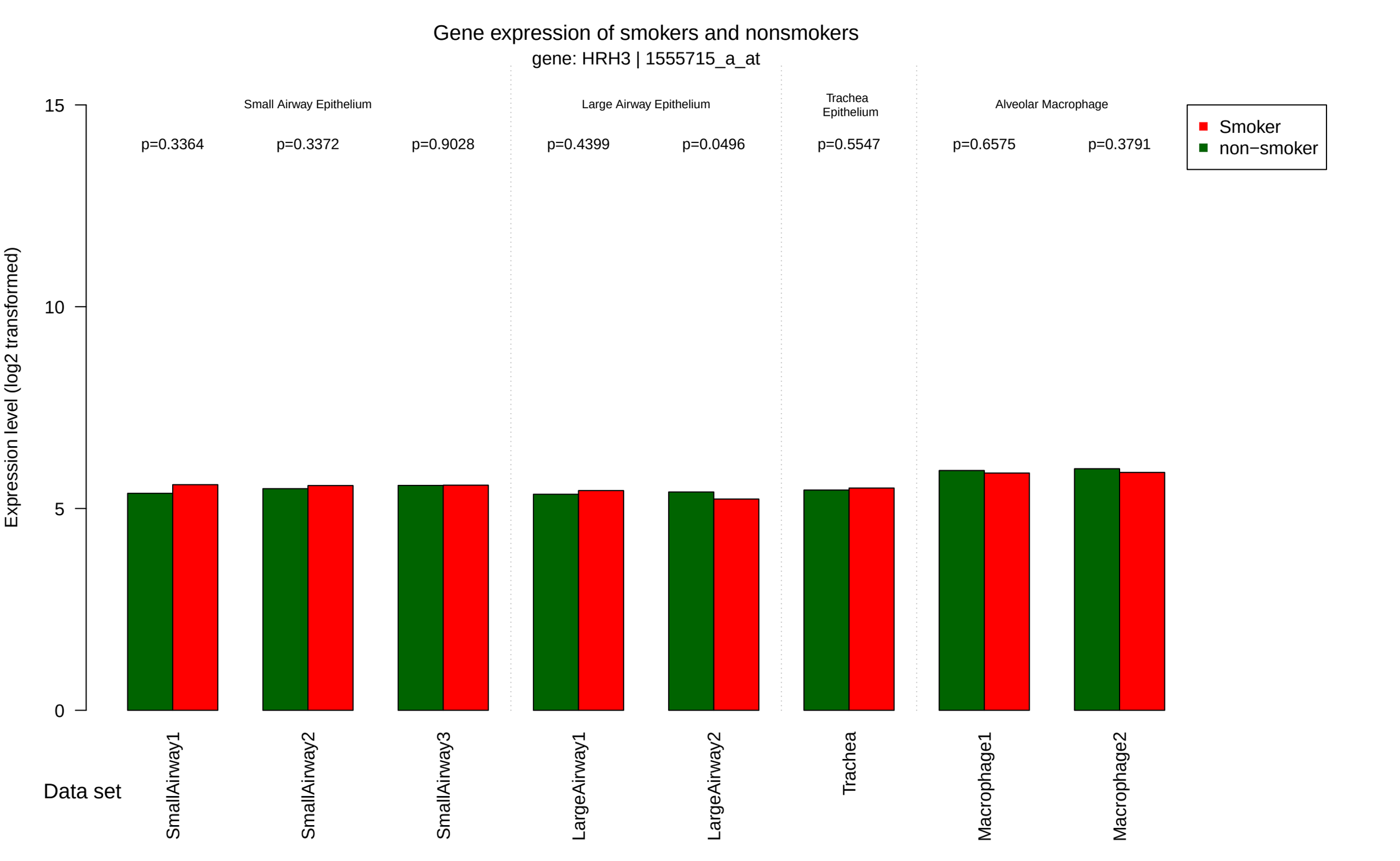
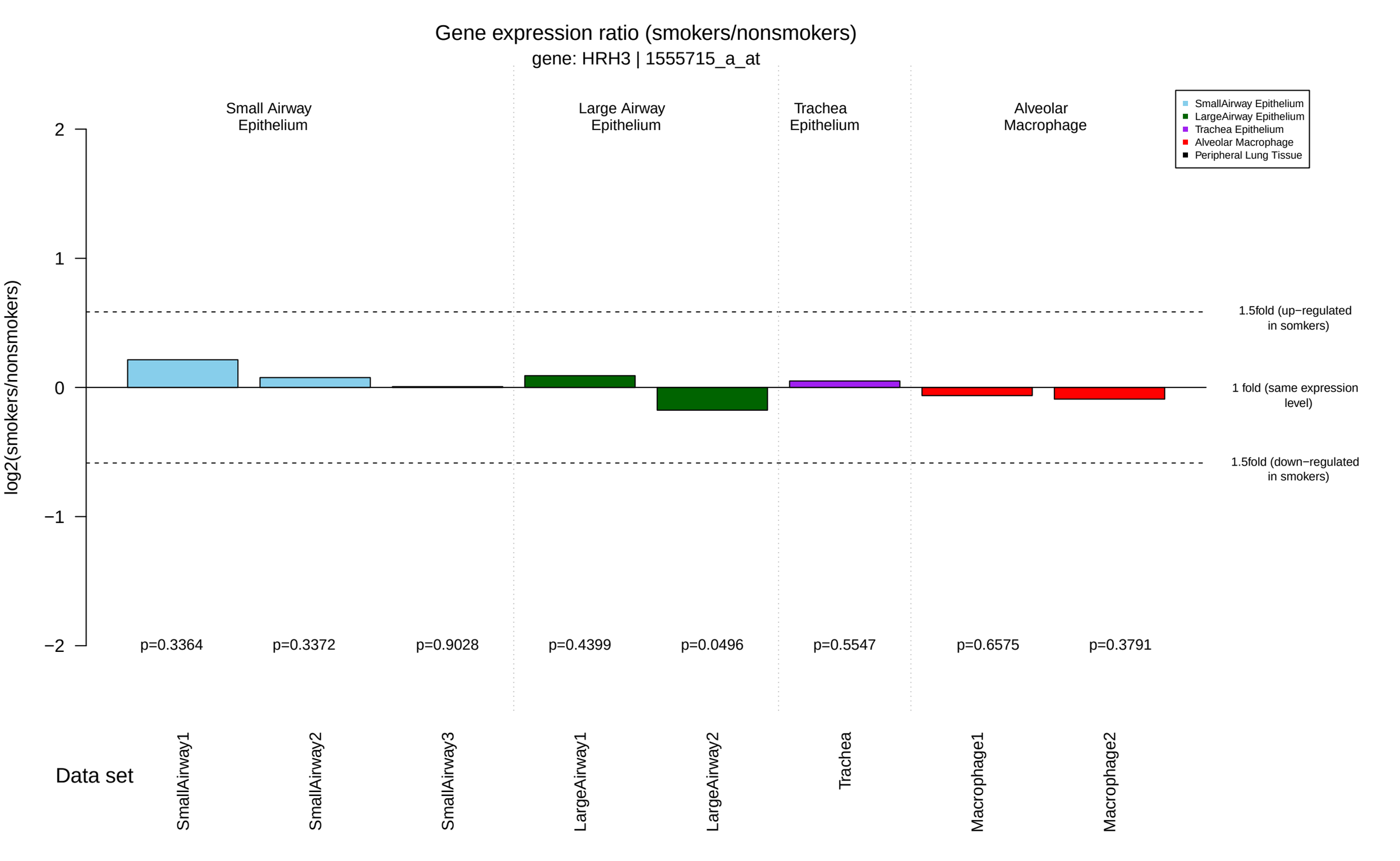
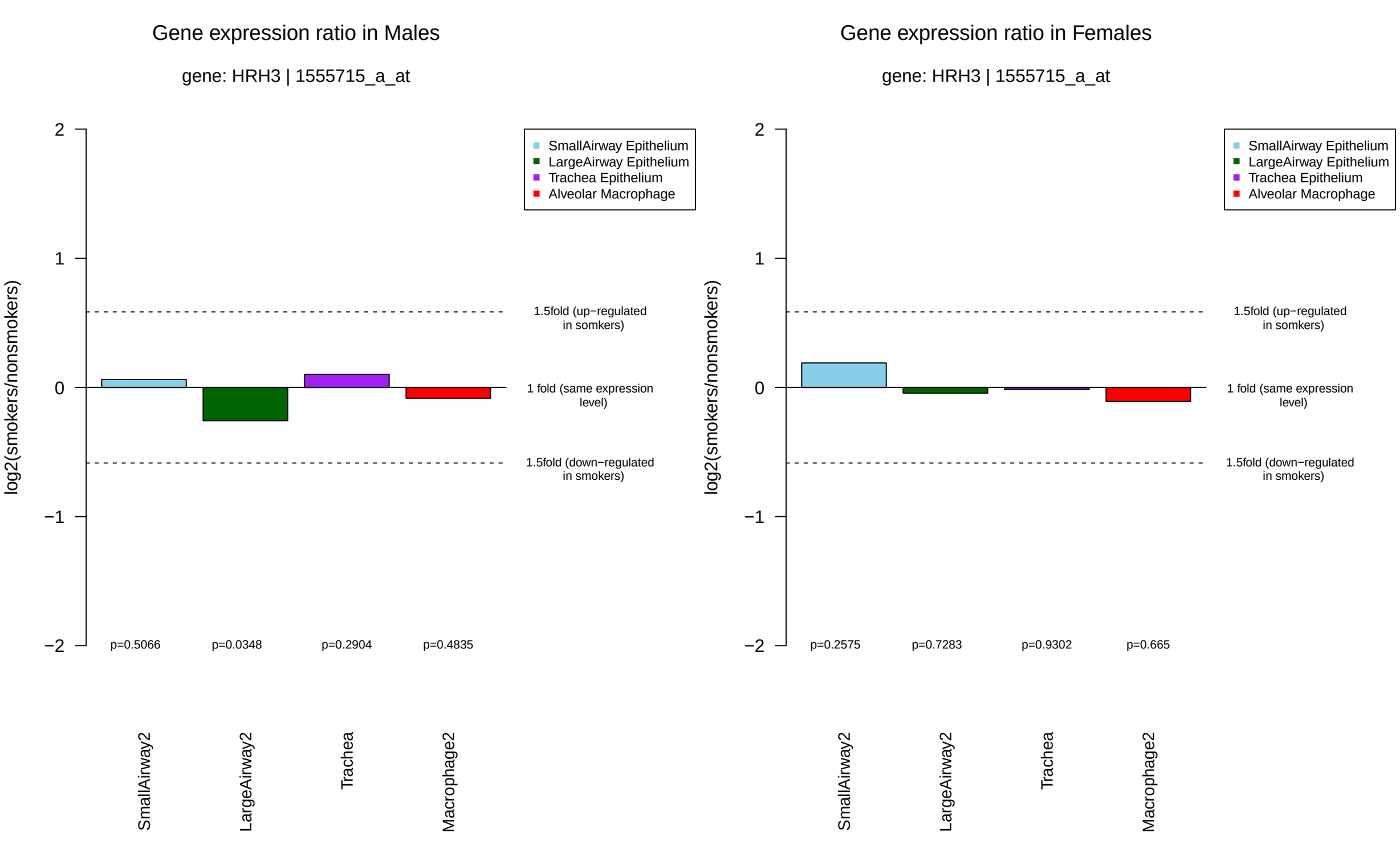
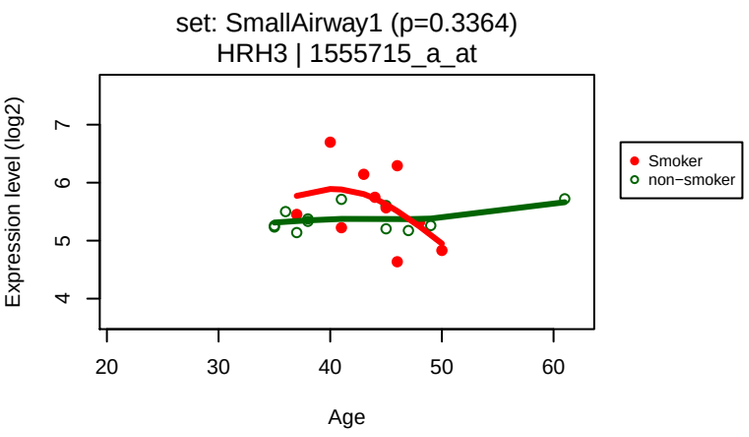
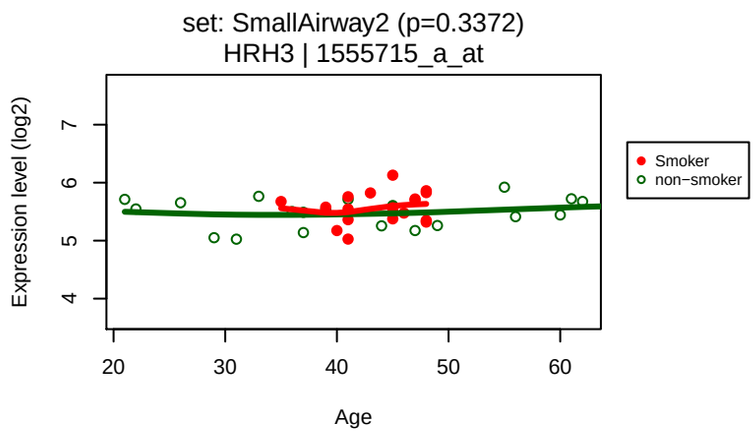
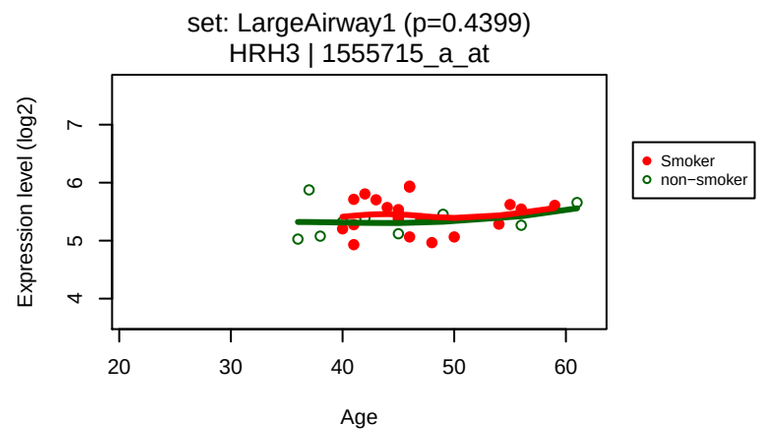

Gene expression of smokers and nonsmokers
gene: HRH3 | 1555715_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3364
p=0.3372
p=0.9028
p=0.4399
p=0.0496
p=0.5547
p=0.6575
p=0.3791
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

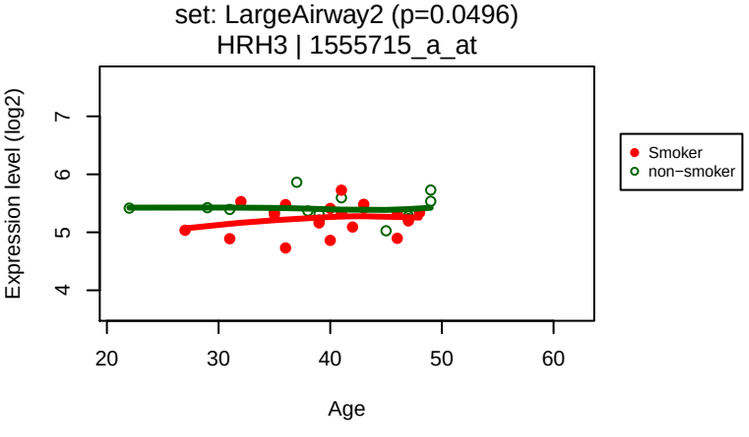

Gene expression ratio (smokers/nonsmokers)
gene: HRH3 | 1555715_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3364
p=0.3372
p=0.9028
p=0.4399
p=0.0496
p=0.5547
p=0.6575
p=0.3791
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

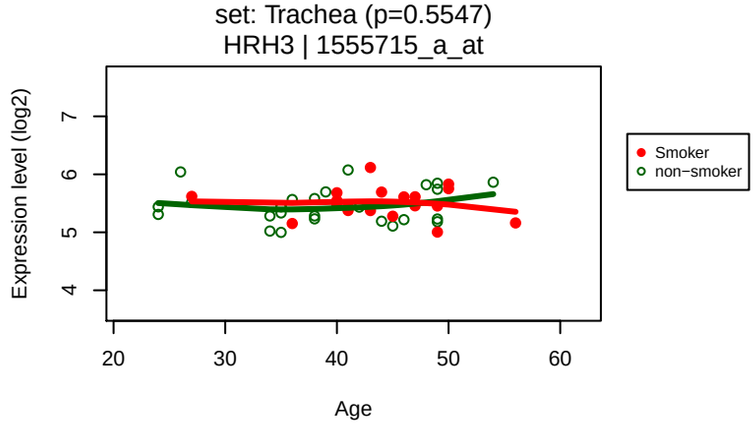

Gene expression ratio in Males
gene: HRH3 | 1555715_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5066
p=0.0348
p=0.2904
p=0.4835
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HRH3 | 1555715_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2575
p=0.7283
p=0.9302
p=0.665
SmallAirway2
LargeAirway2
Trachea
Macrophage2

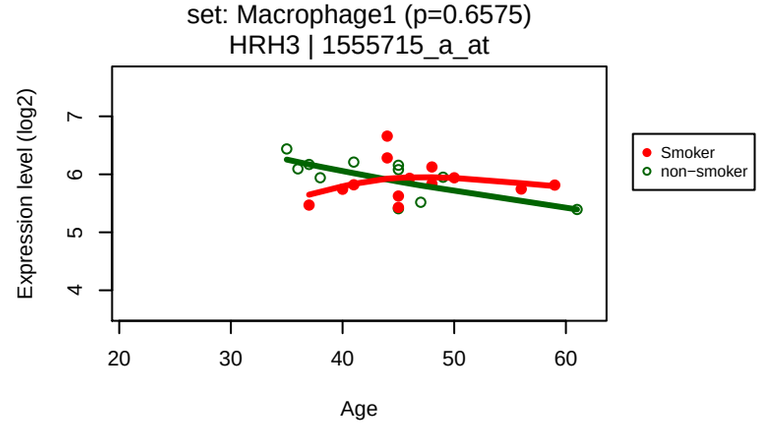

set: SmallAirway1 (p=0.3364)
HRH3 | 1555715_a_at
Expression level (log2)
Age
Smoker
non−smoker

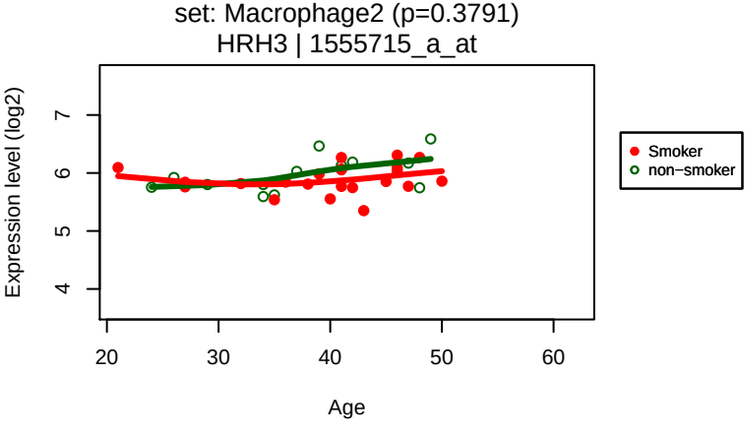

set: SmallAirway2 (p=0.3372)
HRH3 | 1555715_a_at
Expression level (log2)
Age
Smoker
non−smoker

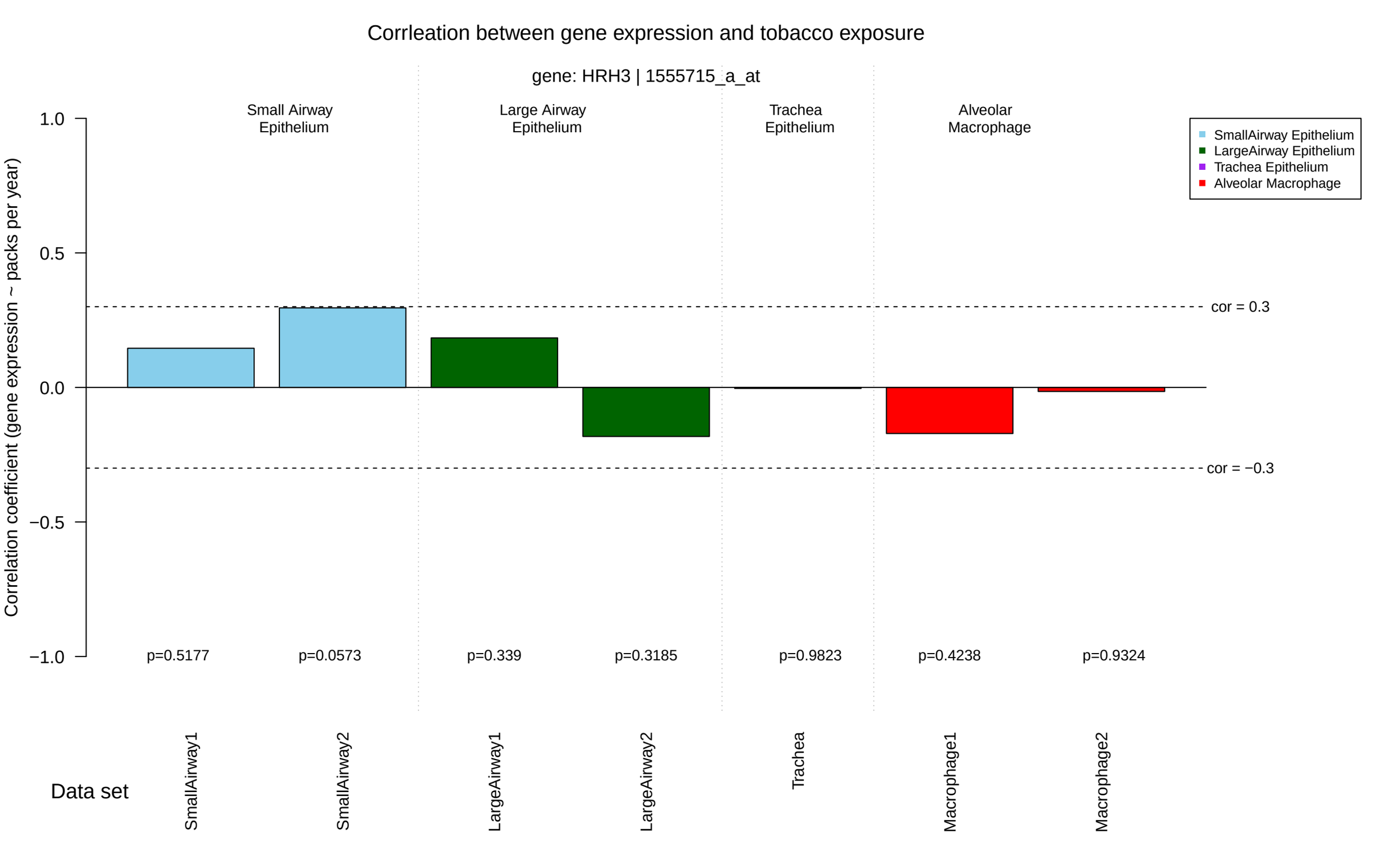

set: LargeAirway1 (p=0.4399)
HRH3 | 1555715_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0496)
HRH3 | 1555715_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5547)
HRH3 | 1555715_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6575)
HRH3 | 1555715_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3791)
HRH3 | 1555715_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HRH3 | 1555715_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5177
p=0.0573
p=0.339
p=0.3185
p=0.9823
p=0.4238
p=0.9324
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2